

ML Depression Diagnosis

Abstract

This project focuses on developing a deep learning-based algorithm for diagnosing depression, aiming to enhance the accuracy and efficiency of mental health assessments. The algorithm identifies key biological, psychological, and social risk factors through an extensive literature review. Machine learning techniques such as Decision Tree, RandomForest, KNN, Gaussian Naive Bayes, and GradientBoostingClassifier will be employed to create predictive models that accurately classify depression. For the Deep Learning techniques, GRU, FCNN, CNN, Transformer, LSTM, and Bert will be employed to create higher accuracy. By integrating multiple variables, the algorithm seeks to improve upon traditional diagnostic methods, which rely heavily on clinical interviews. The final model will be validated using real world social media(Reddit) data to ensure its practical application for early detection and intervention in public health settings.

Introduction

1.1 Background

Depression is a complex and prevalent mental health disorder. Current diagnostic methods rely predominantly on subjective interviews and self-reports, which often lead to inaccurate diagnoses.

1.2 Motivation

With depression affecting over 264 million people worldwide, there is an increasing demand for an objective, data-driven approach to diagnosis, as traditional methods are prone to bias and inconsistencies.

1.3 Objectives

The objective of this project is to develop a deep learning-based algorithm that integrates diverse factors, thereby improving the accuracy, reliability, and predictive power of depression diagnoses.

Data

Reddit(text data)

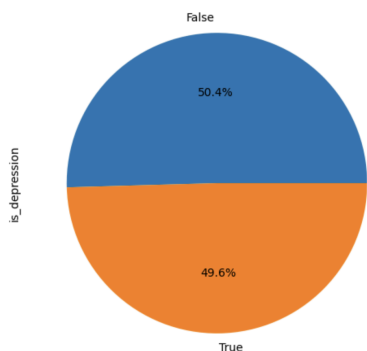


Figure1

Depression is one of the most common mental disorders affecting an individual's personal and professional life. In this work, we investigated the possibility of utilizing social media posts to identify depression in individuals. To achieve this goal, we conducted a preliminary study to extract and analyze the top Reddit posts made in 2022 from depression-related forums. The collected data were labeled as depressive and non-depressive and in Figure 1, it shows that 50.4% of false depression and 49.6% of true depression. The "Depression: Reddit Dataset" has 7,731 entries into a binary column "is_depression," where one means the post is indicative of depression and 0 means the opposite. The dataset is elaborately clean from noise, as special characters and URLs have been removed, rendering perfectly syntactically agreeable text for analysis. It is fitted for sentiment analysis and depression detection tasks; thus, it is a good data set for training and testing machines and deep learning models for depression detection.

Preprocessing

The preprocessing phase involved several steps to prepare the data for analysis. First, stopword removal was applied to eliminate common English stopwords, reducing noise and focusing on meaningful words. Additionally, domain specific stop words such as "http", "www", and "com" were excluded to further clean the data. After that, tokenization was performed using the word_tokenize function from the NLTK library, breaking the text into individual words for detailed analysis. Finally, text cleaning was conducted by removing non-alphanumeric characters and converting all words to lowercase, ensuring consistency and standardization across the dataset. These steps collectively enhanced the quality and relevance of the textual data for subsequent processing.

Word Choice

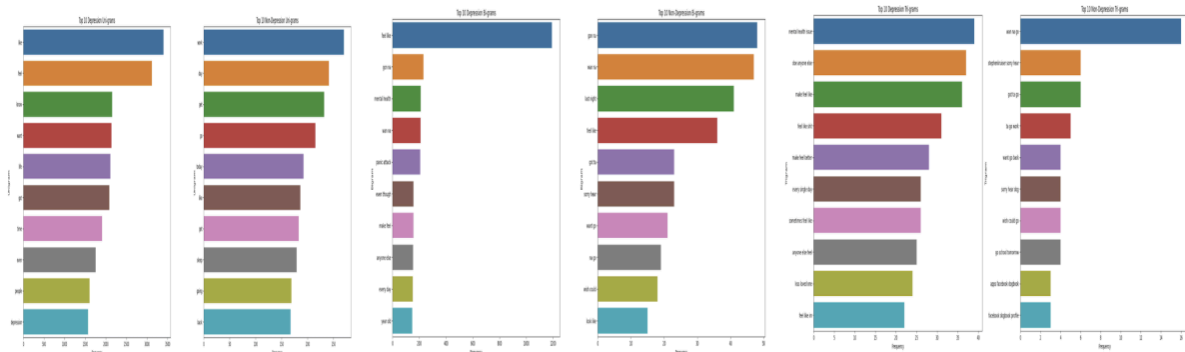


Figure 2

In Figure2, illustrate the frequency of unigrams, bigrams, and trigrams in depression related and non depression related texts providing insight into the linguistic patterns of each category. The n grams were generated using the ngrams function from the NTKL library, allowing for the analysis of single words, word pairs, and word triplets. In depression related texts, unigrams such as "like", "feel" and "know" are frequent, reflecting emotional and

introspective language while non depression texts prominently feature words like “work”, “day”, and “get” focusing on daily activities. Similarly, bigrams in depression texts, such as “feel like” and “mental health”, emphasize personal struggles, whereas non depression texts include casual phrases like “gon na” and “last night”, indicative of routine or conversational topics.

Trigrams further reveal the contrast between the two categories. Depression related trigrams, including “mental health issue” and “make feel like” emphasize discussions around mental health and introspection. Non-depression texts, on the other hand, highlight plans or intentions with trigrams such as “wan na go” and “got ta go”. These patterns underline the emotional and reflective nature of depression related texts compared to the casual and task oriented tone of non depression texts, offering valuable insights into the linguistic markers distinguishing the two.

Textblob

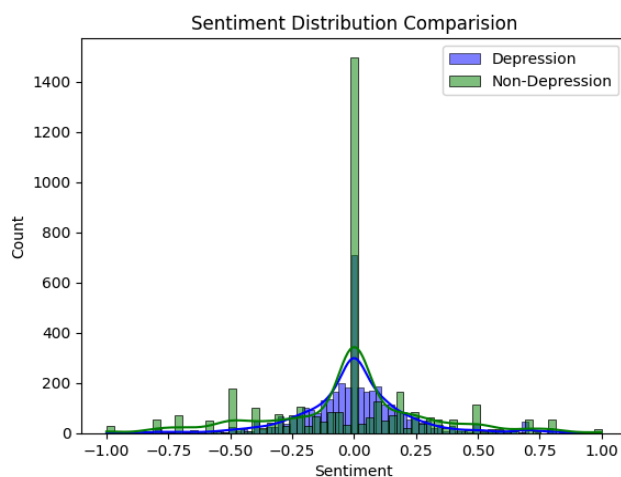


Figure3

In Figure 3, Textblob was used for a shallow sentiment analysis that could analyze the overall sentiment of the texts based on the positive and negative values of words. The sentiment distribution showed distinguishable differences between the two texts, those with and without depression. These differences are pretty fine, and thus one can't define clear boundaries only based on sentiment analysis. This led to the conclusion that probably sentiment alone is not enough to spot depression in text. Next, a machine learning model will be trained on whether a text is depressive or not, which would give a far better and more reliable analysis.

Vectorization

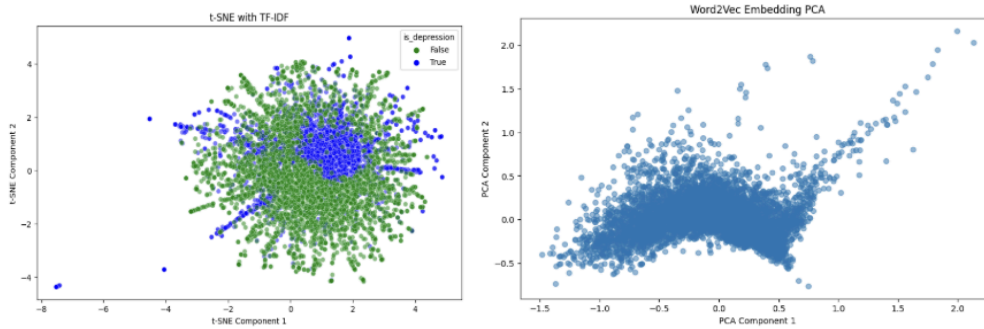


Figure 4

The TF-IDF was used to determine the importance of words within the dataset by converting text into a sparse matrix representation. Each term is weighted with respect to its frequency and importance across documents, hence allowing a numerical representation of text data. To visualize the patterns, t-SNE was applied to the TF-IDF vectors, reducing their high dimensionality to a 2D space. This combination of TF-IDF and t-SNE enables the exploration of the clustering of depression-related versus non-depression text samples. Additionally, the Word2Vec technique was applied to capture semantic relationships between words by representing them as dense vector embeddings in a continuous vector space, reflecting their contextual similarity. Then, PCA was applied to these Word2Vec embeddings to reduce the high-dimensional space to 2D for easier visualization of the semantic relationships.

Models

We employed Machine Learning models: Decision Tree, RandomForest, KNN, Gaussian Naive Bayes, and GradientBoostingClassifier to advocate their particular strengths in depression detection. Decision Tree functions to present a simplified and interpretable baseline; RandomForest for accuracy enhancement and overfitting reduction through ensemble learning. KNN classifies objects with similar characteristics, making KNN design ready for text data. Gaussian Naive Bayes is a quick-learning method that provides a pragmatic probabilistic approach, especially treated as a helpful benchmark. GradientBoostingClassifier delivers a way to identify complex patterns, and it is very efficient with imbalanced data for optimizing classification performance. Diverse model choices will allow us to conduct a thorough evaluation and comparison for depression detection optimization.

Different Models

We have used Deep Learning models, namely GRU, FCNN, CNN, Transformer, LSTM, and BERT, each using their unique strengths for depression detection. GRU captures the sequential dependencies in the text data with efficient modeling of vanishing gradients. FCNN is used as a baseline model, due to its simplicity, and can analyze general patterns. CNN can extract features that are local in representation and hence find intricate patterns within the text. The Transformer model, with its attention mechanism, captures long-range dependencies effectively, ensuring strong context understanding. LSTM can deal with sequential data, having the potential to remember long-term dependencies within the data. Finally, BERT uses bidirectional training with contextual embeddings for better

understanding of the language. These various architectures allow for thorough evaluation and optimization for detecting depression using deep learning techniques.

Experiments

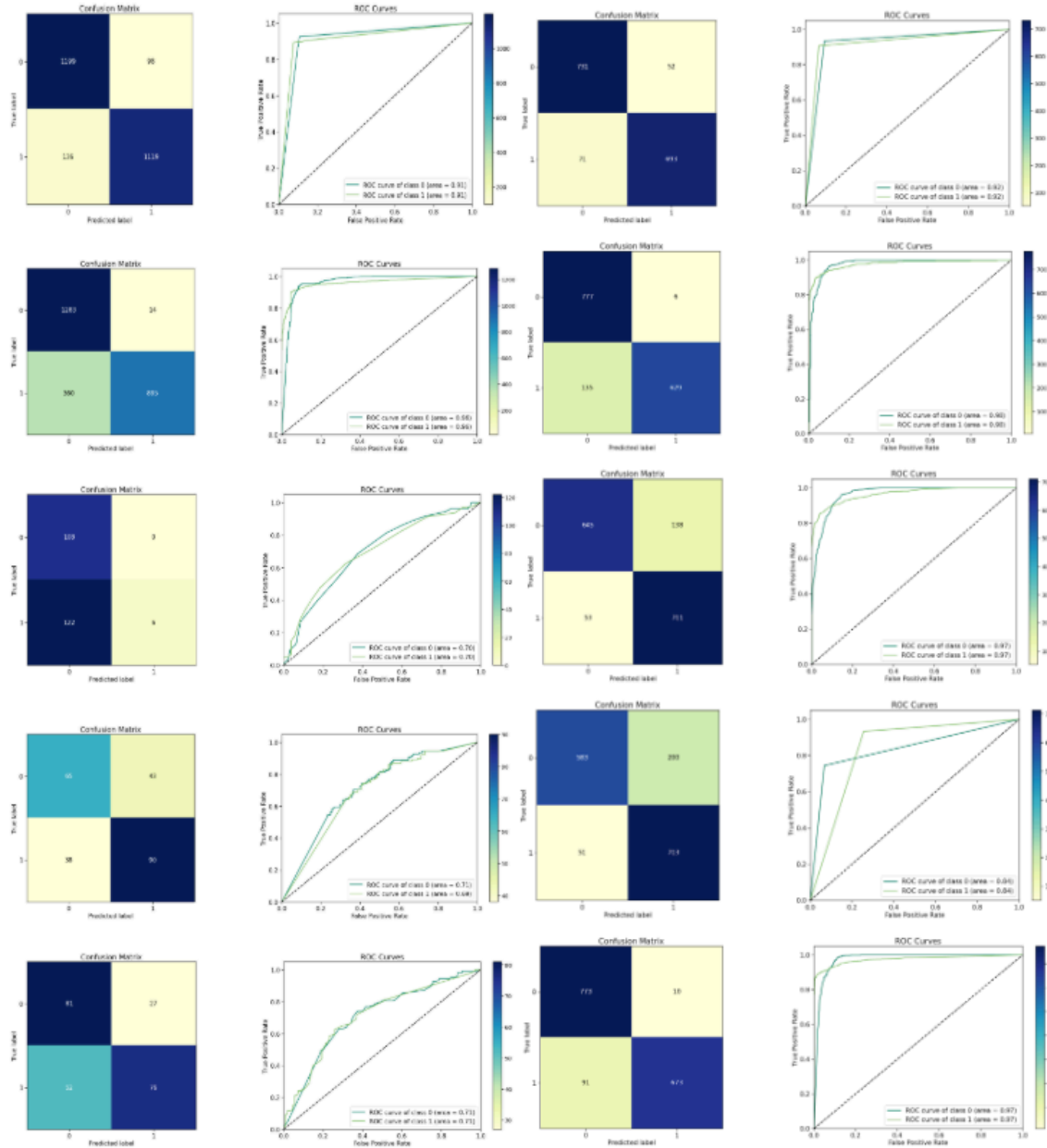


Figure 5

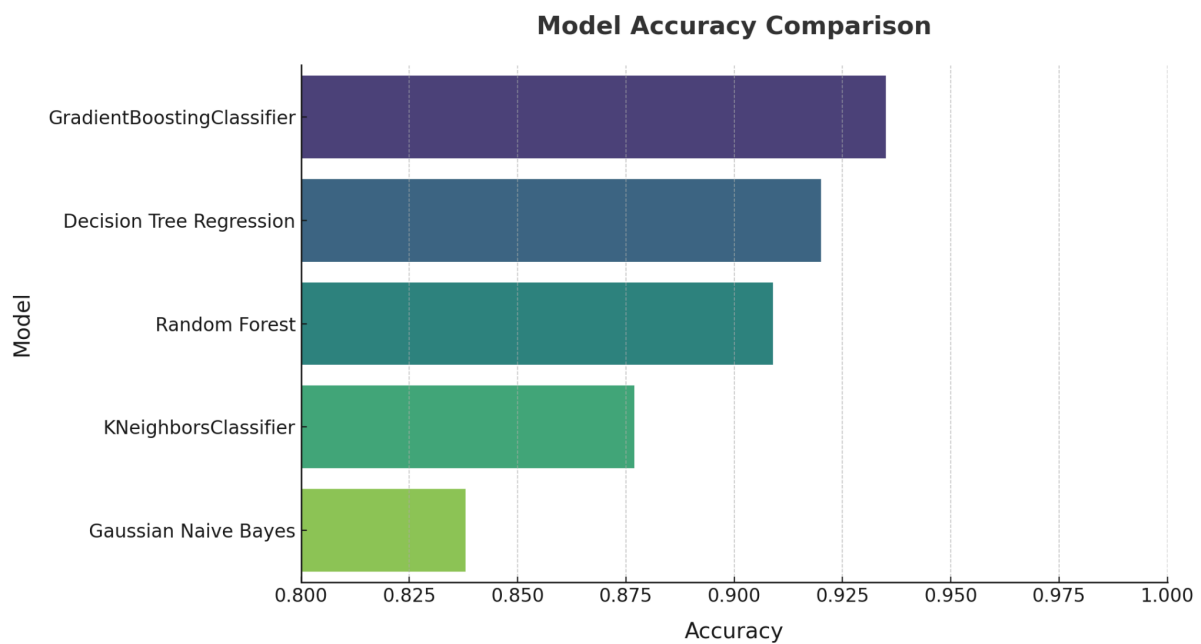


Figure 6

In the figure 5. Vectorization is one of the critical pre-processing steps in machine learning, especially regarding the treatment of textual or categorical data. It involves transforming non-numeric data into a numeric representation, which works correctly for algorithms. It standardizes the data while allowing models to recognize patterns, relationships, and distinctions from the input. For this study, vectorization was applied to structure textual features into numerical representations, including feature vectors or embeddings within the depression-related Reddit dataset. They inform machine-learned models, including Decision Tree Regression, Random Forest, KNeighborsClassifier, Gaussian Naive Bayes (Gaussian Naive Bayes), and Gradient Boosting Classifier, in estimating relations and performing operations at high precision. One of the key advantages vectorization has is that it contains the semantics of information within the textual data. The techniques, like bag-of-words, term frequency-inverse document frequency (TF-IDF), or more advanced methods such as word embeddings, Word2vec, or GloVe, allow for an effective encoding of the features. Those properly encoded features allow the models to develop better generalizing capabilities and make useful predictions based on previously unseen data, a feature especially important in mental health diagnosis, where predictions must have high accuracy and reliability. The other notable feature of vectorization is that it helps speed up computations. In transforming raw, unstructured data into a standardized numeric format, a vectorized dataset facilitates the training and inference phases in the whole pipeline. This becomes especially important when the model is as heavy as a Gradient Boosting Classifier, as one must balance feature scaling and computational resource consumption. Therefore, vectorization is one of the primary building blocks in machine learning workflows. It builds the bridge that connects raw, unstructured data entry into algorithmic processing, allowing models to arrive at structured and meaningful insights during the input phase. In this study, vectorization was similarly critical in deriving insights and improving the applied machine and deep learning models' predictive performance.

As presented in Figure 6, the varying accuracy scores for different machine-learning models applied in the study are shown. GradientBoostingClassifier, with an utmost accuracy score of 0.935, is outstanding in being able to capture non-linear relationships and complicated patterns in the dataset efficiently. This extreme performance implies that enough discretion of the model picks up the best genre of deciding on a depression diagnosis. On the other hand, Gaussian Naive Bayes performed the poorest, with an accuracy score of 0.838, because of its inability to cope with the intricacies of the dataset. The lower accuracy of the model could be related to the fact that it is based on the wrong assumption of feature independence. The results of Figure 5 validate the working of the uncover always behind the shading of the accuracy numbers of several models, thus pointing out how choosing GradientBoostingClassifier becomes vital in tasks concerning quirky and intricate datasets.

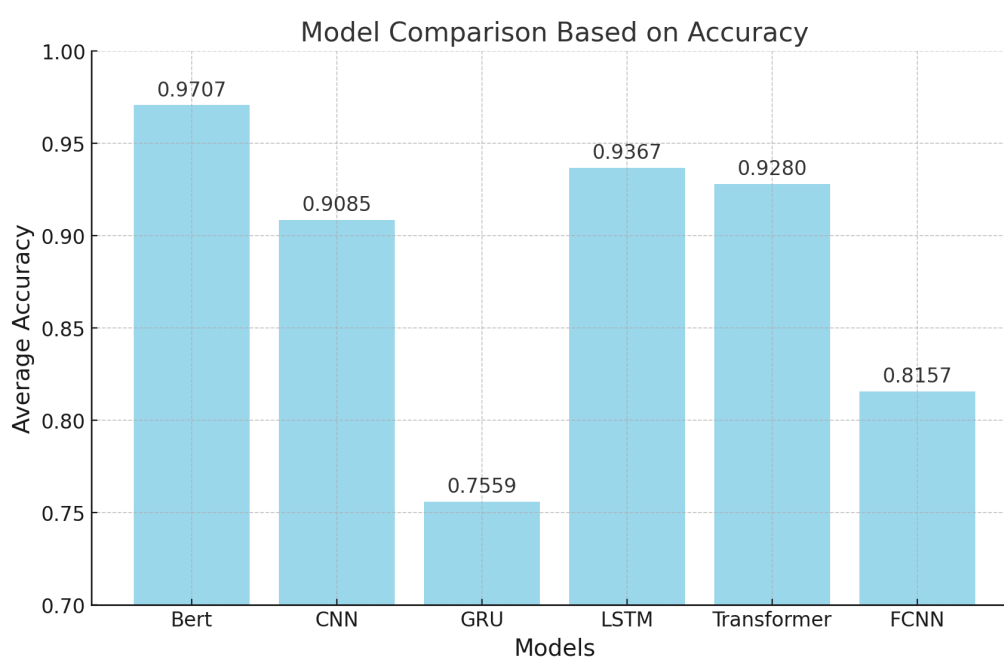
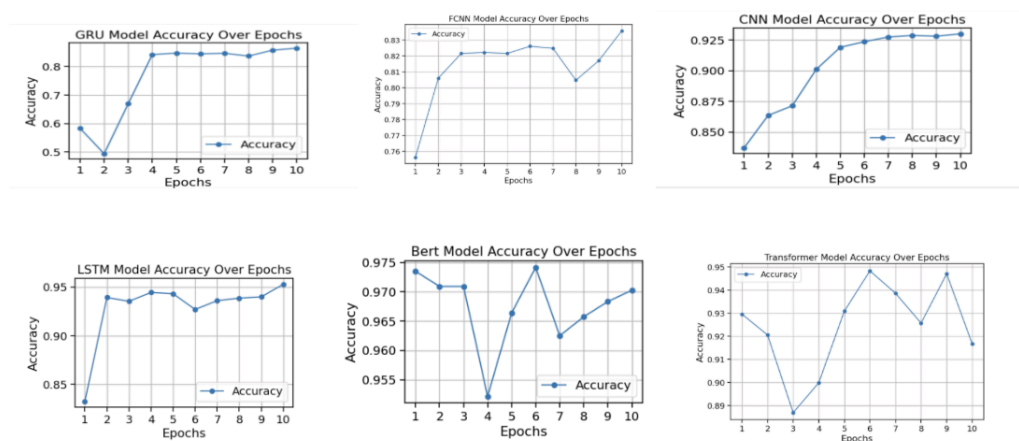


Figure 7 represents the epoch-wise accuracy evolution of various deep learning models that were employed in the optimization of depression diagnosis, such as GRU, FCNN, CNN, Transformer, LSTM, and Bert. Each model is assessed concerning its ability to generalize and capture patterns from depression-related Reddit datasets that underwent complicated preprocessing, such as vectorization, to convert unstructured text into numerical values appropriate for training. It is thus apparent from the figures that Bert, with its transformer-based bidirectional attention mechanism, was always ten steps ahead of the other models, thus achieving peak accuracy on the same set of epochs. LSTM performed real well in facing the substantial problem of long-term dependencies, usually, simpler models such as GRU and FCNN showed comparatively lower accuracy, highlighting the disadvantages associated with processing complex patterns using them. CNN and Transformer models showed fair competition with the hierarchically and attention-based character of their feature extraction capabilities. These findings highlight the importance of selecting advanced architectures like Bert and Transformer for tasks requiring nuanced contextual understanding. The comparative analysis highlighted in the figure presents the strengths and weaknesses of each model, which could give insight into developing robust and accurate tools for mental health diagnosis.

Figure 8 compares the average accuracies of various deep learning models based on the provided code framework. Out of the different models evaluated, the Bert model attained the most remarkable accuracy, an unlikely stellar performance of 0.9707, indicating its superior ability to process the intricate features and patterns within the dataset. This result also displays the advantages of transformer-based architectures like Bert in tasks that require contextual understanding and nuanced representation. On the contrary, GRU scored lowly at 0.7559, indicating it is less suitable for the dataset. This poor performance could result from GRU's simpler architecture compared with models like Bert, which capitalize on advanced attention mechanisms to return superior representations. The results highlight the importance of model selection in achieving optimal performance. While simpler architectures such as GRU may offer faster training, models like Bert provide significantly better accuracy, making them more suitable for high-precision tasks.

Lessons Learned

The project has given us insights into how machine learning and deep learning algorithms can provide and be improved in relation to depression diagnosis. We played with different models to comprehend the respective advantages and disadvantages presented by all the models, particularly in highly complex data sets. From the classical machine-learning models, the greatest accuracy was provided by GradientBoostingClassifier, which was able to recognize complex relationships in the depression-related Reddit dataset, a feature in which the Gaussian Naive Bayes fell flat. For the deep learning portion of the study, the model that performed the best was Bert, again owing to its transformer-based bidirectional

attention mechanisms. GRU, on the other hand, represented the lowest accuracy despite applying to more straightforward sequence-discriminating tasks. This project highlighted the need for model architecture selection for the tasks they are meant to solve. For their capacity for broad generalization when applied to complex situations, GradientBoostingClassifier and Bert performed very satisfactorily. The significance of preprocessing of vectorization was revealed in converting unstructured textual data into forms easily digestible by models for downstream application. This experience sharpened our eyes on various algorithmic behaviors in the same scenario and additional information crucial for the accomplishment of de facto reliable and valid diagnostic tools in the area of mental health.

Conclusion and Future Work

This study optimized various models for the diagnosis of depression with ML and DL algorithms, specifically evaluating and comparing their performances on a depression-related Reddit dataset. This is where we noted numerous differences in the performance of the various models. In GradientBoostingClassifier, considered one of the ML models within the study, this model had the most successful predictive performance, as described by the accuracy feature. It managed to attempt to identify very intricate patterns. On the other hand, as a result of the assumption of the independence of features, it is therefore expected that Gaussian Naive Bayes will lag behind its competitors. In the DL, Bert consistently outperformed the other models due to its bidirectional attention mechanism while the GRU model was not well-suited for the high complexity of the dataset it was working on and managed to achieve very low accuracies. This means one should choose only the advanced architectures, such as GradientBoostingClassifier and Bert, for high-precision and context-dependent tasks. The outcomes also indicated the importance of preprocessing steps like vectorization, crucial in transforming unstructured datasets into forms that the algorithms could process. The comparative review has helped to understand the different strengths and weaknesses of different models and provide a better idea for developing better and more efficient mental health diagnoses. Suggestions for future work would include the following directions to advance the performance and applicability of depression diagnosis models. One is to examine the application of ensemble methods that combine strengths from multiple models, potentially increasing their robustness and accuracy. Including more extensive and more diverse datasets should allow models to generalize better across demographic and linguistic contexts. Adding explainable AI techniques will give these models a degree of interpretability such that their predictions are visible and actionable by mental health professionals. Finally, real-world testing and validation on clinical datasets would narrow the gap between research and practical applications, potentially enabling deployment in real-world contexts.

Reference:

Sarvani, K.; Priya, Y.S.; Teja, C.; Lokesh, T.; Rao, E.B.B. Rainfall Analysis and Prediction Using Machine Learning Techniques. *J. Eng. Sci.* 2021, 12, 531–537